# Lab 8: Combining forecasts

MATH840 | Time Series | Week 9 — graded assignment 7 of 7

AUTHOR
Ihor Miroshnychenko

AFFILIATION
Kyiv School of Economics

**The last weekly lab**

Your series from Week 4, for the fourth time. You have an ETS model from Lab 5 and a seasonal ARIMA from Lab 7, and a published accuracy score for each. Tonight you **put them together** and find out whether that was worth doing — on your series, with your own numbers.

**Open the notebook**

The template: Open in Colab

Click **Copy to Drive** before you start working, otherwise nothing you write is saved.

The practice session — case 4 is this lab on a public series, end to end: Open in Colab · Ukrainian version with results and conclusions

## The task

Forecast your series with an **equal-weight combination** — the *Advanced Forecasting Methods* lecture — and compare it with everything you have built before.

1. **Rebuild** your Lab 5 and Lab 7 models exactly as you submitted them, and check that they reproduce your earlier validation numbers. **Add MSTL** as a third candidate.
2. **Backtest** all three on four rolling origins, next to your benchmark.
3. **Combine** two or three of them — members chosen *before* you read the table — and read the table: would selecting one model have done better?
4. **Choose an interval**: the averaged bounds of the members, or a conformal interval from the combination's own errors — each judged on windows it did not see.
5. **Conclude** against your past results: your Lab 5 and Lab 7 scores on the hidden horizon, and what the combination is expected to add.

**Not this time:** plots of the series, decomposition, ACF and differencing. You have done them three times; nothing in the rubric asks for them again. Every plot tonight is about the models.

**This one counts**

8 points. **Deadline: 23:59 today**, on Moodle. The accuracy score is published the next morning; marks and comments come back before the next practice session.

## What you are marked on

| Section | Criteria | Points |
|---|---|---|
| 1 | Your models, rebuilt — and one new member | 1 |
| 2 | Forecast accuracy against your benchmark | 3 |
| 3 | Justification block | 3 |
| 4 | Code quality and reproducibility | 1 |
| | **Total** | **8** |

The notebook has one section per row. **Keep those headings.**

## 🔁 Section 1 — Your models, rebuilt (1 point)

- `MEMBERS` holds your **submitted** Lab 5 ETS and Lab 7 seasonal ARIMA — the specification, the damping, the log — and MSTL, with a log decision of its own.
- `PAST` holds the hidden-horizon MASE and skill you were sent after Labs 5 and 7.
- The template prints each member's MASE on the validation window. **It must match your earlier notebooks.** If it does not, find out why — a different specification, `AutoETS` choosing differently on a refit — and say so.
- Two or three sentences: do the numbers match, and where does MSTL land next to your two models?

## 🎯 Section 2 — Forecast accuracy (3 points)

What you submit is **an equal-weight average of two or three members**, with an 80% interval from one of the two recipes. No single model, no estimated weights, no model families beyond the three.

Scored from your submission file on the `h` observations after your history that nobody has seen:

$$\text{skill} = \frac{\text{MASE of your forecast}}{\text{MASE of your benchmark}}$$

| Skill | Points |
|---|---|
| below 1 — you beat your benchmark | 3 |
| 1 to 1.1 — within 10% | 2 |
| above 1.1 | 1, and 3.3 must say why |

> ⓘ **What the pool guarantees, and what it does not**
>
> Measured on all 140 issued series, with **automatic** members fitted on the raw history — `AutoETS`, `AutoARIMA`,

`MSTL` — and scored on the hidden horizon:

| Forecast | Beats the benchmark | Within 10% | Worse | Median skill | Worst skill |
|---|---|---|---|---|---|
| `AutoETS` + `AutoARIMA` | **100%** | — | — | 0.805 | 0.93 |
| all three | 96% | 3% | 1% | 0.798 | 1.18 |
| select the best of three on four windows | 96% | 2% | 2% | 0.811 | 1.23 |
| `MSTL` alone | 80% | 6% | 14% | 0.812 | 2.46 |

- The pool was screened so that `AutoETS` and `AutoARIMA` each beat the benchmark — so their average does too. **MSTL was not screened**: on one series in seven it loses by more than 10% on its own, and it can pull a combination with it. Whether it belongs in yours is a decision for 2.2.
- Your Lab 5 and Lab 7 models may not be the automatic ones. The table is about the automatic ones.
- **Intervals, on the hidden horizon:** the averaged bounds cover 78% of the months on average, the conformal recipe 79%; Winkler scores 3.75 and 3.85. Neither recipe wins in general — which is why you choose on your own windows.

## ✍ Section 3 — Justification block (3 points)

The part a human reads closely. Roughly a paragraph each, with numbers from your own cells.

| Item | Points | What earns them |
|---|---|---|
| 3.1 | 1 | **Combine or select.** The members you chose *before* the table and why each is defensible on its own. The rolling table read: who won how many windows, where the combination landed in each, how it compares with the best single member. The error correlations — how much there was to gain. What selection would have submitted, and why you combined anyway. |
| 3.2 | 1 | **The interval.** Both recipes on unseen windows: coverage, Winkler score, width. The one you submitted and why. How many errors the conformal radius rests on — four windows of $h$ — and what that means for trusting it. |
| 3.3 | 1 | **Conclusions against your past results.** Your Lab 5 and Lab 7 hidden-horizon skills next to the rolling evidence for the combination: did combining help on your series, against which model, by how much? The skill you expect tonight, and why. One thing you would now do differently in Lab 5 or Lab 7. |

**Important**

Marked on the reasoning. A combination that does no better than your best model scores full marks in 3.1 if you show it with numbers and say why you would still submit it — or why you would not. A rebuild that does not match, honestly explained, costs nothing.

⚠ **Where this goes wrong**

- **Choosing the members after reading the table.** Then the combination is a selection in disguise — the noisy decision it was meant to avoid. Write 2.2's note first.
- **Comparing on the validation window alone.** One window is one draw; the practice session's tourism case showed selection on three windows picking the test winner barely more often than a coin.
- **Judging the conformal interval on the windows it was built from.** It will look perfect. The template's `interval_check` leaves each window out in turn — use it.
- **Averaging forecasts on different scales.** Every member's forecast must be on the original scale — `fit_member` takes care of the log; do not exponentiate twice.
- **Forecasting from `train`.** The submission refits every member on the whole history.

📖 FPP3 §13.4 Forecast combinations · §5.5 Distributional forecasts and prediction intervals · FPP3 §12.1 Complex seasonality (MSTL)

## 🖌 Section 4 — Code quality and reproducibility (1 point)

- The notebook runs top to bottom on a fresh kernel without errors.
- The submission file is written **by the notebook**, not edited by hand.
- No leakage: members and interval recipes are compared on the rolling windows; only the final members are fitted on all of `y`.
- Every number in your text comes from a cell.

## 📤 Submission

Upload three files to the Week 9 activity on Moodle **before 23:59 today**:

- `SURNAME_lab09_forecast.csv` — written by the last cell of the template. Exactly $h$ rows and these columns:

| Column | What |
|---|---|
| `student_id` | your identifier |
| `code` | your series code, the same as before |
| `ds` | the dates you are forecasting, in order |
| `yhat` | the combination's point forecast |
| `yhat_lo_80`, `yhat_hi_80` | the 80% interval you chose |

- `SURNAME_lab09.pdf` — the report, with the justification block;
- `SURNAME_lab09.ipynb` — the source.

> ⚠ **A file that cannot be read scores 0 for accuracy**
>
> Exactly $h$ rows, the dates asked for, no `NaN`, the interval around the point forecast. The template's check cell is the same one the scorer uses — if it passes, your file is fine.

## ⏱ How long this should take

Ninety minutes. The backtest refits three members on four origins — on a monthly series with a seasonal ARIMA, a few minutes; run it once and keep `bt`. Most of the time should go into Section 3.

## 🔭 What comes next

This was the last weekly lab. Keep this notebook: next week's lecture, *Foundation Models for Time Series*, asks whether a pretrained network beats the combination you submit tonight.